Project Planning

Software Project Planning

The overall goal of project planning is to establish a pragmatic strategy for controlling, tracking, and monitoring a complex technical project.

OR

A Plan is the strategy for the successful completion of the project. It's a description of the project steps that produce increasing maturity of the products or processes produced by the project.

Why?

So the end result gets done on time, with quality!

Reasons people don't plan

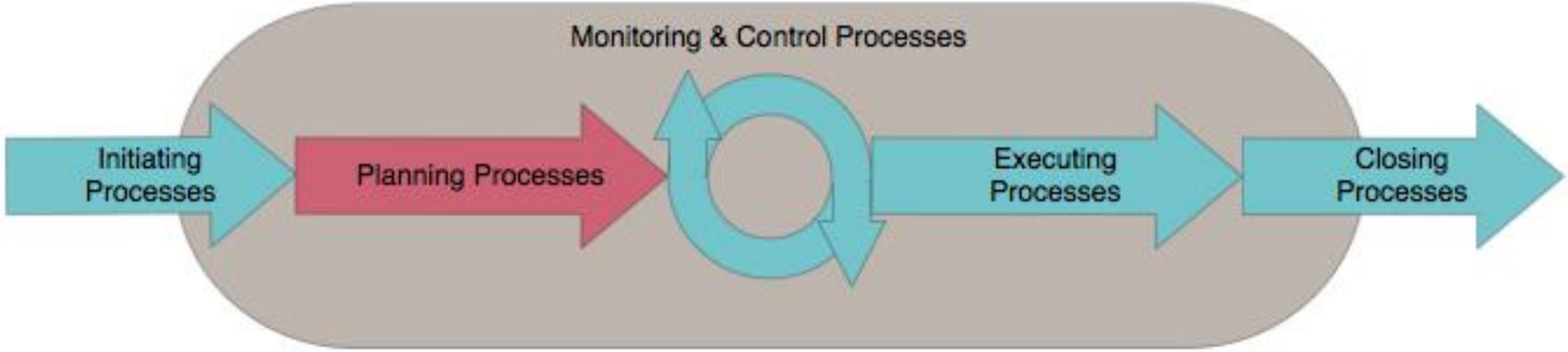
- Don't believe in planning
  - Management or organization culture does not support planning
- Find the process painful
  - If you do plan, the pain will be greatest early and will diminish with time
  - If you don't plan, you defer the pain (and it will usually be greater…)
- Don't have time to plan!
  - Consider the simple task of getting yourself to class
  - Now consider having the responsibility of getting all the other people to class on time

Planning is essential to control

- An effective way to exert control is to:
  - Know where you are
  - Know where you are supposed to be
  - Take corrective action if there is a difference between the two
- Note:
  - You have to have a plan to know where you are supposed to be
  - If you have no plan, you have no control
    - *Example*: Commercial airliner flying from Chicago to Tokyo
- Planning and control
  - Two phases both needed

Planning processes

- **Planning processes** determine the total project scope, define or refine the project objectives, and develop the course of action to achieve the objectives
- Planning employs *progressive elaboration*: the process of revisiting planning (and possibly initiating) processes as additional project information becomes available
- The planning processes encompass project integration management, project scope management, project time management, project risk management, and project stakeholder management



Planning

- **Scoping**
  - What is the problem?
- **Estimation**
  - How much will it cost?
- **Schedule**
  - How long will it take?
- **Resources**
  - How many people will it take?
- **Risk**
  - What might go wrong?
- **Control Strategy**

Planning processes and knowledge areas

Knowledge Area Processes	Planning Process Group
Project Integration Management	• Develop Project Management Plan
Project Scope Management	• Plan Scope Management • Collect Requirements • Define Scope • Create WBS
Project Time Management	• Define Activities • Sequence Activities • Estimate Activity Resources • Estimate Activity Durations • Develop Schedule
Project Risk Management	• Identify Risks • Perform Qualitative Risk Analysis
Project Stakeholder Management	• Plan Stakeholder Management

Primary Planning Steps

1. Identify project scope and objectives
2. Define and Record Requirements
3. Identify project organizational environment
   - Analyze project characteristics
   - Identify Project Team and Define Roles and Responsibilities
4. Identify project products and activities
   - Create the WBS
   - Estimate effort for each activity
   - Allocate resources
   - Schedule deliveries and milestones

Primary Planning Steps

5. Develop Change Management Plan
6. Identify Risks and Define Risk Strategies
7. Review and communicate plan
8. Obtain Plan Approval
9. Conduct Kick-off Meeting

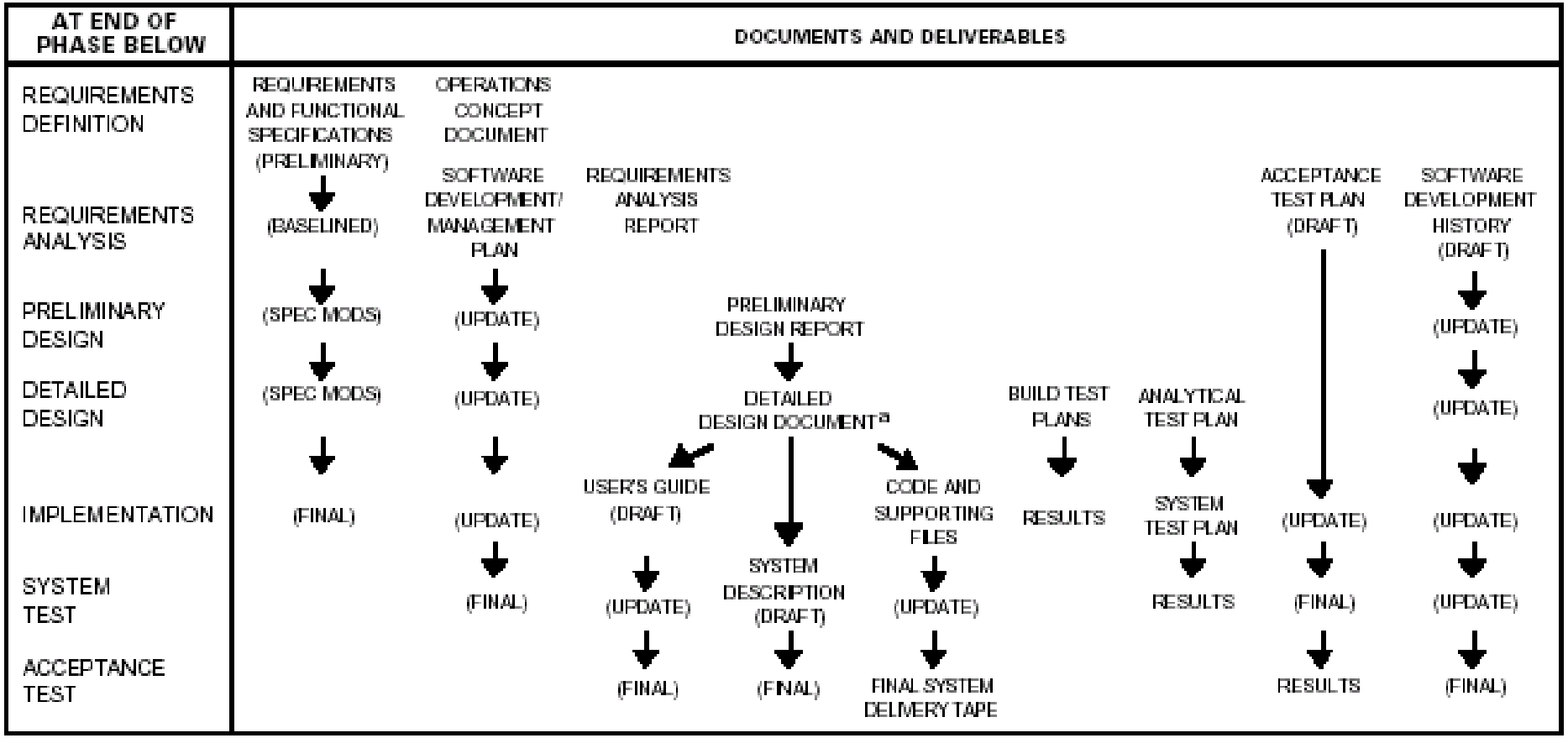
Project Planning Task Set – I

- Establish project scope
- Determine feasibility
- Define required resources
  - Determine required human resources
  - Define reusable software resources
  - Identify environmental resources
- Estimate cost and effort
  - Decompose the problem
  - Develop two or more estimates using size, function points, process tasks or use-cases
  - Reconcile the estimates

Project Planning Task Set – II

- Develop a project schedule
  - Scheduling is considered in detail later.
    - » Establish a meaningful task set
    - » Define a task network
    - » Use scheduling tools to develop a timeline chart
    - » Define schedule tracking mechanisms
- Analyze risks
  - Risk analysis

Plans Evolve Over Time



NASA's “Manager's Handbook for Software Development”

Software Development Plan (SDP)

Software Project Management Plan (SPMP)

- Some consider it the most important document in the project (along with requirements document)
  - Can be seen as an aggregation of other core documents
- Evolves over time as pieces come together

SDP / SPMP

Fundamental Sections

- Project overview
- Deliverables
- Project organization
- Managerial processes
  - Communication management plan
  - Milestones
- Technical processes
- Budget
- Schedule

Deliverables

- List of items to be delivered
  - Must be tangible items
- Sample deliverables
  - The product – the actual software, in the installable format
  - Product documentation
  - Reports and planning documentation
- Most projects are driven by deliverables, so you need several
- **Project** deliverables (aka documents) are included

Communications Management Plan

- Often a section of Software Project Management Plan (SPMP)
- Describes information flow to all parties
  - Gathering and distributing information
- Status meetings
  - Monthly, Weekly, Daily?
  - Status reports are vital
- Stakeholder Management Plan

Planning Documents

- Project ROI Analysis
  - Business Case (include competitive analysis if appropriate)
- Project Charter
  - Statement of Work (SOW)
- Software Project Management Plan (SPMP)
  - Software Development Plan (SDP)
    - Schedule
  - Communications Management Plan
- Budget
- Responsibility Assignment Matrix (RAM)
- Risk Management Plan

Example of Responsibility Assignment Matrix (RAM)

Following is an example of a responsibility assignment matrix for a new product launch project.

Task	Responsible	Accountable	Consulted	Informed
Conduct market research	Marketing team	Product manager	Sales team	Executive team
Design product packaging	Design team	Product manager	Marketing team	Executive team
Source materials for production	Procurement team	Product manager	Design team	Executive team
Set up production line	Operations team	Product manager	Procurement team	Executive team
Create marketing campaign	Marketing team	Product manager	Sales team	Executive team

Planning Documents

Other documents you may want/need

- Software Quality Assurance Plan (SQAP)
  - Software Process Improvement Plan
- Software Configuration Management Plan (SCMP)
- Migration Plan
- Operations Plan

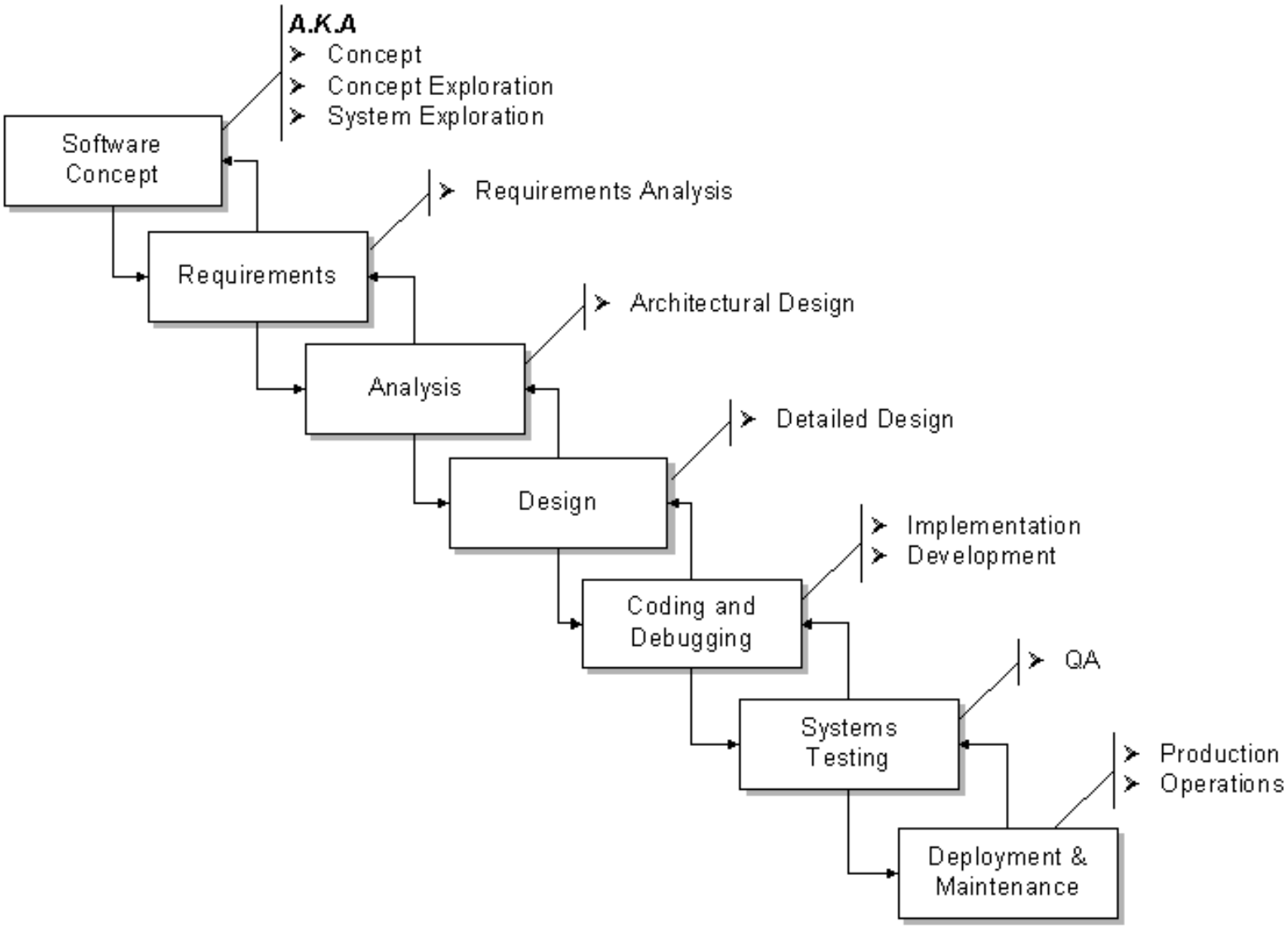
Planning Documents

- You (the PM) need to choose which documents are appropriate
- Docs do not have to be lengthy
- Small Set:
  - Software Development Plan
  - Risk Management Plan
  - Software Quality Assurance Plan
  - Software Configuration Management Plan

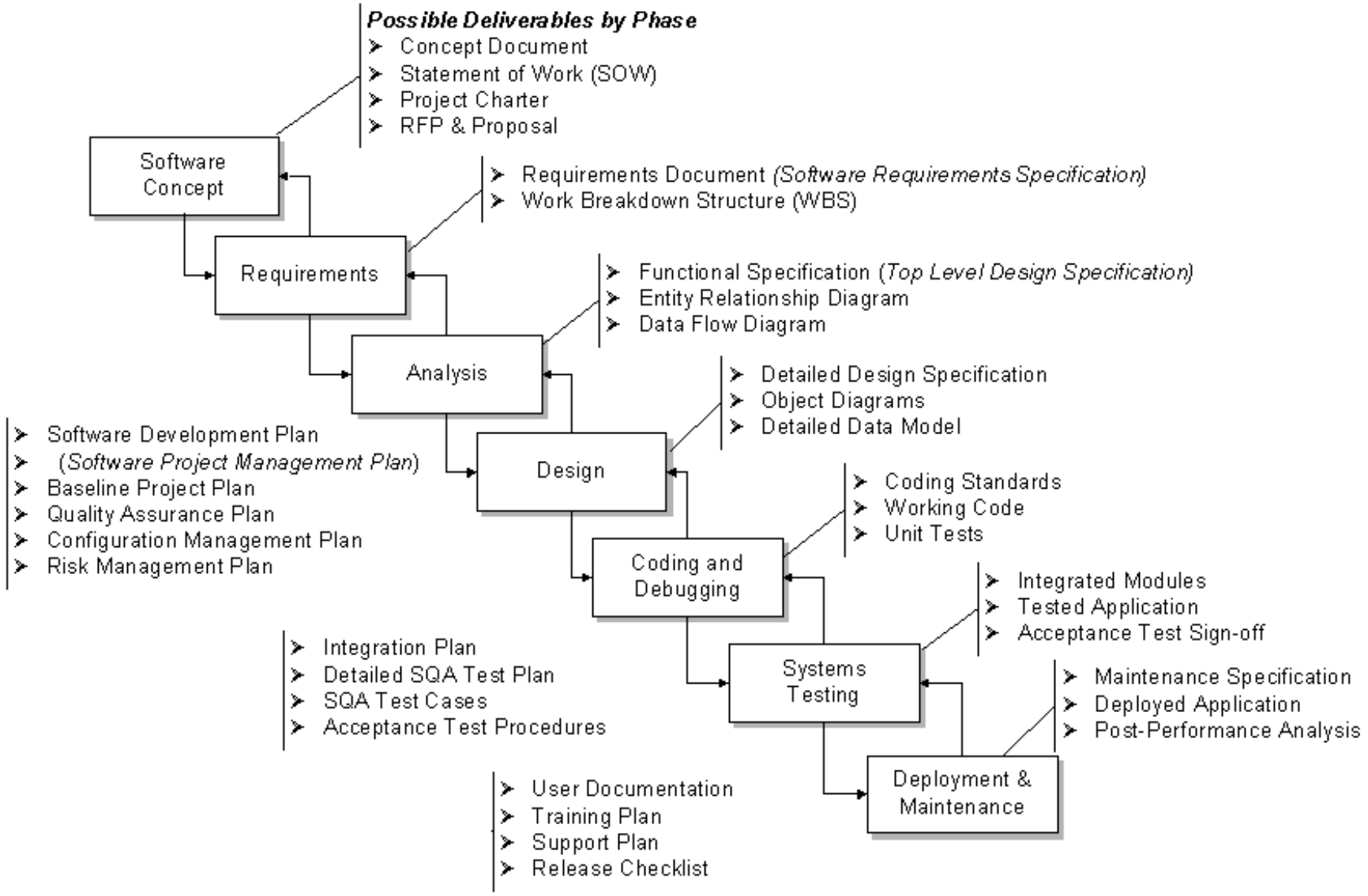
Product Documents

- Statement of Need
- System Interface Specification
- Software Requirements Specification
- Software Design Specification
- Software Validation & Verification Plan
- User Documentation
- Support Plan
- Maintenance Documentation

Project Phases



Potential Deliverables by Phase



Time Allocation by Phase

- Remember the 40-20-40 Rule
  - Specification-Implementation-Test

	Planning	Code & Unit Test	Integration & Test
Commercial DP	25%	40%	35%
Internet Systems	55%	15%	30%
Real-time Systems	35%	25%	40%
Defense Systems	40%	20%	40%

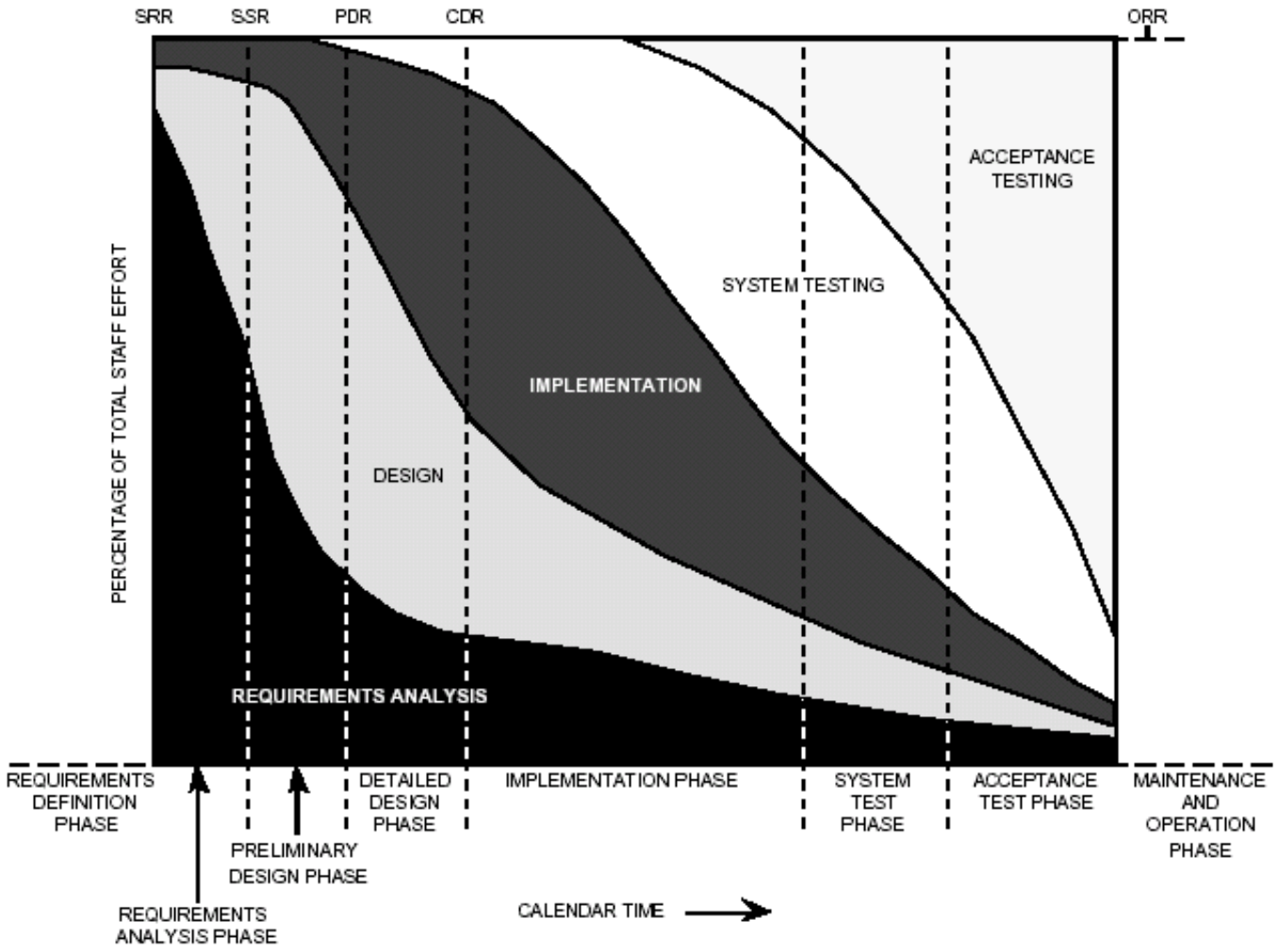
Bennatan, E.M, "On Time Within Budget"

Time Allocation by Phase

Activity	Small Project (2.5K LOC)	Large Project (500K LOC)
Analysis	10%	30%
Design	20%	20%
Code	25%	10%
Unit Test	20%	5%
Integration	15%	20%
System test	10%	15%

McConnell, Steve, “Rapid Development”

Activities by % of Total Effort



NASA's “Manager's Handbook for Software Development”

Conventional planning approach

- Planning seems to be a straightforward process:
  - Determine the tasks to be done
  - Determine the order of the tasks
  - Execute the tasks in the proper order
- Uncertainty can, and usually does, disrupt this flow
  - Not every task can be identified before the project starts
  - Contingent factors—internal or external—add, delete, or modify tasks
  - Task ordering is changed due to new or unforeseen dependencies
- At their very foundations, conventional planning approaches implicitly assume an unrealistic model of human cognition and behavior, while ignoring or underestimating real-world risk

Planning in an IID methodology

- In contrast to big, up-front planning or even planning in conventional iterative and incremental methodologies, an IID methodology spreads planning out over the entire project lifecycle
- Planning in an IID methodology:
  - Distributes planning from the top-level stakeholders to the individual developers—each plans at the appropriate scale
  - Delays fine-grained planning to 'just-in-time' before corresponding task or activity
  - Is low-overhead— 'barely sufficient' planning documents are lightweight and the majority are produced 'as-needed' rather than as a matter of protocol
  - Is resilient to requirements changes—it embraces change rather than attempts to prevent or avoid it